

Nguyen Thanh Tien

Intern Backend / IoT Engineer

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Overview

I am a fourth-year Software Engineering student at Ton Duc Thang University, currently seeking an Intern Backend or IoT Engineer position. I have experience in building high-concurrency, decoupled systems using Node.js/Express, MQTT, message queue (Redis/BullMQ). My expertise includes developing robust backend architectures, managing real-time data flow, and integrating low-latency embedded solutions with ESP32. I am committed to applying my technical skills to professional projects that require scalable and efficient software solutions.



Education

Aug 2022 – Now

Ton Duc Thang University

Major - Software Engineering



Skills

- Main**
- JavaScript (Node.js/Express.js), Java, C/C++ (ESP32).
 - IoT & Protocols: ESP-NOW, MQTT, WebSocket (Socket.IO).
 - SQL, NoSQL (MongoDB), Redis, BullMQ (Queue).
 - Server (Nginx, Docker, Docker Compose), Log, GitHub Actions.
 - Firebase (Authentication, Firestore, Realtime Database, Storage, Messaging).
 - English: Upper-Intermediate (B2) - Aptis ESOL Certificate (Issued by British Council)
- Other**
- GIT
 - UML Modeling
 - Basic Proficiency in UI Design Tool (Figma)
 - Work Management Tool (Jira)
 - Familiarity with Agile and Scrum principles



SCALABLE SYSTEM DESIGN & DEPLOYMENT

(10/2025)

Source code	https://github.com/TDeV-VN/midterm-nodejs-tdtu
Descriptions	A project dedicated to learning and professionally applying Docker in a real-world context, focusing on designing and deploying a robust, scalable backend system. It emphasizes modern DevOps practices and tooling for a streamlined development and production workflow.
Number of members	1
Position	Backend & DevOps Engineer
Responsibilities	System Architecture: • Designed a decoupled system where services (API, Worker) can be scaled horizontally by replicating container instances. • Implemented a Redis-based message queue (BullMQ) to decouple the API from background workers, ensuring reliable asynchronous task processing and improving system resilience. DevOps & Containerization: • Orchestrated the entire system using Docker Compose, creating distinct, isolated environments for development and production. • Authored optimized multi-stage Dockerfiles to produce lean, secure production images, drastically reducing their size and potential vulnerabilities. • Configured Nginx as a reverse proxy and a secure entry point to the system. Professional Workflow & Tooling: • Managed application settings for different environments (dev/prod) through centralized configuration files and environment variables. • Integrated a suite of professional tools to enhance the development lifecycle: Nodemon for hot-reloading, ESLint for code quality enforcement, and Dozzle for real-time log aggregation and viewing across all running containers.
Technology in use	• Architecture: Scalable Design, Service Decoupling, Message Queues (Redis, BullMQ) • DevOps: Docker, Docker Compose, Nginx, Environment Configuration • Developer Tooling: Dozzle (Log Aggregation), ESLint, Nodemon • Backend: Node.js, Express.js, MongoDB

E-COMMERCE WEB PLATFORM (GEAR UP)

(11/2025 - 12/2025)

Source code	https://github.com/TDeV-VN/tdtu-nodejs-ecommerce
Descriptions	A full-featured E-commerce system with a multi-service architecture. The platform includes a customer-facing website (Next.js), an admin dashboard (React/Vite), a central backend API (Node.js/Express), and a background worker for asynchronous tasks.
Number of members	3
Position	Backend & DevOps Engineer
Responsibilities	System Architecture & Backend: - Contributed to developing a decoupled system where services (API, Worker, Frontend) operate independently. - Implemented a Redis-based message queue (BullMQ) to decouple the API from background workers, ensuring reliable asynchronous task processing (e.g., sending emails) and improving system resilience. - Developed and maintained RESTful APIs for features: user authentication, shopping cart, orders, and a discount system. - Integrated real-time features using Socket.IO for live product reviews, enhancing user interaction. DevOps & Development Workflow: - Orchestrated the entire system using Docker Compose, creating distinct, isolated environments for development and production. - Authored optimized multi-stage Dockerfiles to produce lean, secure production images, reducing their size and potential vulnerabilities. - Managed application settings for different environments through centralized configuration files. - Collaborated within the team using Git and enforced code quality standards with ESLint.
Technology in use	- Architecture: Service-Oriented Architecture, Service Decoupling, Message Queues (Redis, BullMQ), Real-time (Socket.IO). - Backend: Node.js, Express.js, MongoDB, Mongoose, Passport.js. - Frontend: Next.js, React.js, Redux Toolkit, Vite, Tailwind CSS, TypeScript. - DevOps & Tooling: Docker, Docker Compose, Git, ESLint.

Emergency Vehicle Preemption System (EVPS)

(10/2025 - 12/2025)

Source code	https://github.com/TDeV-VN/EVPS-PublicPreview.git
Descriptions	An advanced IoT ecosystem designed to grant priority to emergency vehicles at traffic intersections. The system leverages low-latency device-to-device communication, real-time tracking, and a centralized management dashboard to optimize emergency response routes.
Number of members	2
Position	System Architecture & Backend & Embedded Engineer
Responsibilities	IoT & Embedded Systems (Core Focus): - Designed a low-latency P2P communication system using ESP-NOW. - Secured wireless data transmission by implementing AES-128 ECB encryption (mbedtls) for ESP-NOW payloads, preventing unauthorized traffic light manipulation. - Processed raw NMEA data from NEO-M8N modules using TinyGPS++ and implemented the Haversine formula for precise distance-based traffic light triggering. - Developed OTA (Over-The-Air) firmware update mechanisms integrated with GitHub Releases for automated deployment. Backend & System Architecture: - Built a high-concurrency backend with Node.js/Express, processing real-time MQTT telemetry and broadcasting updates to clients via Socket.IO. - Decoupled background tasks (email alerts, system syncing) using BullMQ and Redis to improve system reliability. - Containerized the entire microservices architecture with Docker for seamless and scalable deployment.
Technology in use	- IoT & Embedded: ESP32 , C/C++, ESP-NOW, Wokwi (Simulation). - Protocols: MQTT, WebSocket (Socket.IO), RESTful API, HTTP/HTTPS. - Backend: Node.js, Express.js, Redis, BullMQ (Task Queue), MongoDB. - Frontend: React.js, Vite, Tailwind CSS, Leaflet.js (Maps). - DevOps & Tools: Docker & Docker Compose, Nginx, GitHub Actions (CI/CD), Git, Jira.